Day 120 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Unit 8 Test

Standards: 6.RP.A.3c, 6.G.A.1, 6.G.A.3, 6.G.A.4, 6.G.A.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

Assessment routine

5 minutes: explain response types without solving any item. 35 minutes: independent assessment. 10 minutes: collect work and note misconceptions for the next lesson. Keep the answer key and worked-example help unavailable during assessment.

Review allocation

Questions 1–4 are previous-unit review (20%). Current-unit questions increase in demand after the review. Unit 1 uses Grade 5 readiness skills because no earlier course unit exists.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2, 4, 19, 23

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Directions

The first 4 questions (20%) review Unit 7. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Write 25% as a simplified fraction and decimal.

02 / Previous unit review | Constructed response

Write 10% as a simplified fraction and decimal.

03 / Previous unit review | Constructed response

Write 290% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

04 / Previous unit review | Constructed response

Find 50% of 100.

05 / Constructed response

Sketch a triangle with base 12 cm and perpendicular height 6 cm. Find its area.

06 / Constructed response

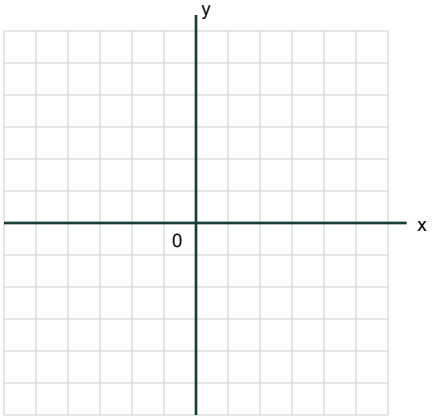
A parallelogram has base 5 m and perpendicular height 2 m. Sketch how it can be rearranged into a rectangle and find its area.

07 / Constructed response

An L-shaped garden is a 4 m by 7 m rectangle with an 2 m by 5 m corner rectangle removed. Sketch and find the garden area.

08 / Constructed response

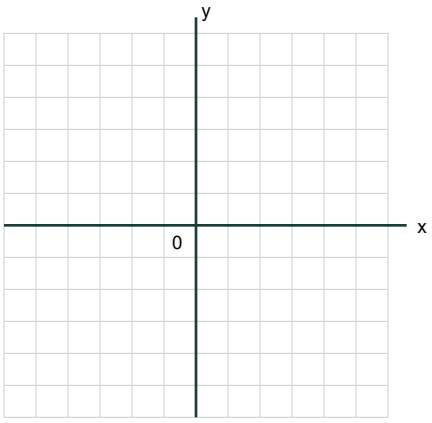
Plot and connect (-5,-5), (2,-5), (2,2), (-5,2). Find the side lengths and area.



Label the scale and units.

09 / Constructed response

Plot and connect (-6,-3), (5,-3), (5,5), (-6,5). Find the side lengths and area.



Label the scale and units.

10 / Constructed response

Draw a net for a closed 4 × 7 × 5 cm rectangular prism. Label the six faces and calculate total area.

11 / Constructed response

A 5 cm by 6 cm base and 5 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

12 / Net reasoning and numeric entry

A square pyramid net has base side 12 cm and four triangular faces, each with perpendicular face height 10 cm. Part A: enter one triangular face area. Part B: enter the entire surface area.

13 / Numeric entry and model reasoning

A prism measures 9/2 m by 1 m by 5/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

14 / Two-part numeric entry

A closed box is 6 cm by 4 cm by 5 cm. Part A: enter its surface area. Part B: the 6-by-4 top is removed; enter the remaining surface area and explain.

15 / Numeric entry and model reasoning

A prism measures 11/2 m by 3/2 m by 5/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

16 / Numeric entry and model reasoning

A prism measures 7/2 m by 1 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

17 / Two-part numeric entry

A closed box is 5 cm by 6 cm by 3 cm. Part A: enter its surface area. Part B: the 5-by-6 top is removed; enter the remaining surface area and explain.

18 / Two-part numeric entry

A 10 m by 11 m rectangle has a right-triangle corner removed with perpendicular legs 5 m and 6 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

19 / Multiple select

A triangle has base 8 cm and perpendicular height 3 cm. Select all expressions that give its area in cm².
A. 4 × 3
B. (8 + 3) ÷ 2
C. (8 × 3) ÷ 2
D. 8 × 3

20 / Numeric entry and error analysis

A trapezoid has parallel bases 6 m and 14 m, and perpendicular height 3 m. Part A: enter its area. Part B: a student multiplies the longer base by the height; explain why that overcounts.

Day 120 | Unit 08 | Answer key

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1. 1/4; 0.25

25% = 25/100 = 1/4; divide by 100 for 0.25.

6.RP.A.3c | Constructed response

2. 1/10; 0.1

10% = 10/100 = 1/10; divide by 100 for 0.1.

6.RP.A.3c | Constructed response

3. 29/10; 2.9

290% = 290/100 = 29/10. Dividing by 100 gives 2.9, which exceeds 1 because 290 exceeds 100.

6.RP.A.3c | Constructed response

4. 50

50/100 × 100 = 50.

6.RP.A.3c | Constructed response

5. 36 cm²

A = bh/2 = 12 × 6 ÷ 2 = 36. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

6. 10 m²

Cut and shift a corner; A = bh = 5 × 2 = 10.

6.G.A.1 | Constructed response

7. 18 m²

Outer area 28 minus cutout 10 = 18.

6.G.A.1 | Constructed response

8. Sides 7 and 7; area 49 square units

Horizontal length 2 - (-5) = 7; vertical length 2 - (-5) = 7; area = 7 × 7.

6.G.A.3, 6.G.A.1 | Constructed response

9. Sides 11 and 8; area 88 square units

Horizontal length 5 - (-6) = 11; vertical length 5 - (-3) = 8; area = 11 × 8.

6.G.A.3, 6.G.A.1 | Constructed response

10. 2 faces each of 4×7, 4×5, 7×5; 166 cm²

Add 2(28) + 2(20) + 2(35) = 166.

6.G.A.4 | Constructed response

11. 170 cm²

Closed surface area = 2(30+25+30) = 170.

6.G.A.4 | Constructed response

12. Face 60 cm²; total 384 cm²

One face = (12 × 10)/2 = 60. Four faces total 240. Add square base area 144.

6.G.A.4 | Net reasoning and numeric entry

13. 45/4 m³; 90 cubes

Multiply edges: 9/2 × 1 × 5/2 = 45/4. The edge counts are 9, 2, 5, giving 90 cubes. Each cube is 1/8 m³; 90/8 = 45/4.

6.G.A.2 | Numeric entry and model reasoning

14. 148 cm²; open box 124 cm²

Closed area = 2(24+30+20) = 148. Subtract exactly one top face, 24, for the open box.

6.G.A.4 | Two-part numeric entry

15. 165/8 m³; 165 cubes

Multiply edges: 11/2 × 3/2 × 5/2 = 165/8. The edge counts are 11, 3, 5, giving 165 cubes. Each cube is 1/8 m³; 165/8 = 165/8.

6.G.A.2 | Numeric entry and model reasoning

16. 21/4 m³; 42 cubes

Multiply edges: 7/2 × 1 × 3/2 = 21/4. The edge counts are 7, 2, 3, giving 42 cubes. Each cube is 1/8 m³; 42/8 = 21/4.

6.G.A.2 | Numeric entry and model reasoning

17. 126 cm²; open box 96 cm²

Closed area = 2(30+15+18) = 126. Subtract exactly one top face, 30, for the open box.

6.G.A.4 | Two-part numeric entry

18. Removed 15 m²; remaining 95 m²

Outer area = 110. Triangle = 5 × 6/2 = 15. Subtract the removed region exactly once.

6.G.A.1 | Two-part numeric entry

19. A, C - 4 × 3; (8 × 3) ÷ 2

Triangle area is half the matching parallelogram: (8 × 3)/2 = 12 cm². Multiplying half the base by the height gives the same result.

6.G.A.1 | Multiple select

20. 30 m²

Use average base × height: (6 + 14)/2 × 3 = 30. A full rectangle at the longer base includes extra corner regions.

6.G.A.1 | Numeric entry and error analysis